

T2A1: Derive the fourth-order Runge–Kutta algorithm

The problem

Solve $dy/dt = g(y, t)$ with $y(0) = y_0$. If I know $y(t)$, I want to find $y(t + \tau)$.

The idea

The exact answer is the Taylor series:

$$y(t + \tau) = y + \tau y' + \frac{\tau^2}{2} y'' + \frac{\tau^3}{6} y''' + \frac{\tau^4}{24} y^{(4)} + O(\tau^5)$$

with $y' = g$ and, using the chain rule (Eq. 4.22 in the book),

$$y'' = g_t + g g_y, \quad y''' = g_{tt} + 2g g_{ty} + g^2 g_{yy} + g_y(g_t + g g_y).$$

These derivatives of g are annoying to compute. So instead I take a weighted average of g at a few points in the step. The weights and points are chosen so the result matches the Taylor series up to τ^4 .

Warm-up: two slopes

First I try two slopes (Eqs. 4.25–4.32 in the book):

$$y(t + \tau) = y + \alpha_1 c_1 + \alpha_2 c_2, \quad c_1 = \tau g(y, t), \quad c_2 = \tau g(y + \nu c_1, t + \nu \tau).$$

Taylor expanding c_2 :

$$c_2 = \tau g + \nu \tau^2 (g_t + g g_y) + O(\tau^3).$$

Putting this back in:

$$y(t + \tau) = y + (\alpha_1 + \alpha_2) \tau g + \alpha_2 \nu \tau^2 (g_t + g g_y) + O(\tau^3).$$

Comparing with the Taylor series:

$$\alpha_1 + \alpha_2 = 1, \quad \alpha_2 \nu = \frac{1}{2}.$$

This is 2 equations for 3 unknowns, so there is a free choice, e.g. $\alpha_1 = \alpha_2 = \frac{1}{2}$, $\nu = 1$. For fourth order I do the same, just with more terms.

Four slopes

Now I use four slopes (Eqs. 4.23–4.24 in the book):

$$y(t + \tau) = y + \alpha_1 c_1 + \alpha_2 c_2 + \alpha_3 c_3 + \alpha_4 c_4$$

$$c_1 = \tau g(y, t)$$

$$c_2 = \tau g(y + \nu_{21} c_1, t + \lambda_2 \tau)$$

$$c_3 = \tau g(y + \nu_{31} c_1 + \nu_{32} c_2, t + \lambda_3 \tau)$$

$$c_4 = \tau g(y + \nu_{41} c_1 + \nu_{42} c_2 + \nu_{43} c_3, t + \lambda_4 \tau)$$

where λ_i is the sum of the ν 's in row i . There are 10 unknowns: 4 α 's and 6 ν 's.

I expand each c_i like in the warm-up and compare with the Taylor series up to τ^4 . Each type of term gives one equation, 8 in total:

$$\begin{aligned} \tau^1 : \quad & \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 = 1 \\ \tau^2 : \quad & \sum \alpha_i \lambda_i = \frac{1}{2} \\ \tau^3 : \quad & \sum \alpha_i \lambda_i^2 = \frac{1}{3}, \quad \alpha_3 \nu_{32} \lambda_2 + \alpha_4 (\nu_{42} \lambda_2 + \nu_{43} \lambda_3) = \frac{1}{6} \\ \tau^4 : \quad & \sum \alpha_i \lambda_i^3 = \frac{1}{4}, \quad \text{plus 3 more for the mixed terms} \end{aligned}$$

8 equations and 10 unknowns leaves 2 free choices.

Choosing the parameters

I take the slope at the start, twice in the middle and at the end: $\lambda_1 = 0$, $\lambda_2 = \lambda_3 = \frac{1}{2}$, $\lambda_4 = 1$. Then:

$$\begin{aligned} \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 &= 1 \\ \frac{1}{2}(\alpha_2 + \alpha_3) + \alpha_4 &= \frac{1}{2} \\ \frac{1}{4}(\alpha_2 + \alpha_3) + \alpha_4 &= \frac{1}{3} \end{aligned}$$

Subtracting the last two gives $\alpha_2 + \alpha_3 = \frac{2}{3}$. Therefore $\alpha_4 = \frac{1}{6}$ and $\alpha_1 = \frac{1}{6}$.

Next I choose $\alpha_2 = \alpha_3 = \frac{1}{3}$. The simplest ν 's that satisfy the other equations are $\nu_{21} = \nu_{32} = \frac{1}{2}$, $\nu_{43} = 1$ and the rest zero. Check: $\frac{1}{3} \cdot \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{6} \cdot 1 \cdot \frac{1}{2} = \frac{1}{6}$. The remaining equations also hold.

Result

$$y(t + \tau) = y(t) + \frac{1}{6}(c_1 + 2c_2 + 2c_3 + c_4)$$

$$c_1 = \tau g(y, t)$$

$$c_2 = \tau g(y + \frac{1}{2}c_1, t + \frac{1}{2}\tau)$$

$$c_3 = \tau g(y + \frac{1}{2}c_2, t + \frac{1}{2}\tau)$$

$$c_4 = \tau g(y + c_3, t + \tau)$$

This is Eqs. 4.33–4.37 in the book. Starting from y_0 , I repeat this step. The error per step is $O(\tau^5)$.

Discussion

There are more unknowns than equations, so the parameters are not unique. The classical choice is the simplest, since most ν 's are zero. Another option is the points $0, \frac{1}{3}, \frac{2}{3}, 1$, which gives Kutta's 3/8 rule: $y(t + \tau) = y + \frac{1}{8}(c_1 + 3c_2 + 3c_3 + c_4)$. The book also says the parameters can be adjusted to the problem to improve accuracy.